REPORT

ARTIFICIAL INTELLIGENCE (CSA1704)

Assessment Tool 1 – Case Study

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LOGICAL REASONING AND KNOWLEDGE-BASED AGENTS

INTRODUCTION

Artificial Intelligence (AI) systems can use knowledge representation and logical reasoning to make decisions from available facts and rules. This assessment focuses on four important AI applications: a Smart Medical Diagnosis System, an Autonomous Traffic Management Agent, an Agricultural Expert Reasoning System, and a Wumpus World Knowledge Agent. The problems demonstrate the use of propositional logic, First-Order Logic (FOL), unification, forward chaining, backward chaining, CNF conversion, and resolution.

CASE STUDY 1 – SMART MEDICAL DIAGNOSIS SYSTEM

Problem Statement

A hospital is designing a rule-based logical agent for preliminary patient diagnosis. The system uses symptoms and medical rules to identify possible diseases.

Knowledge Base

1. If a patient has Fever AND Cough → Possible Flu.
2. If a patient has Fever AND Rash → Possible Measles.
3. If a patient has Cough AND Breathlessness → Possible Pneumonia.

Facts about Patient A

- Fever = True
- Cough = True
- Breathlessness = True

(a) Propositional Logic Representation

Let:

- F = Fever
- C = Cough

- R = Rash
- B = Breathlessness
- Flu = Flu
- Measles = Measles
- Pneumonia = Pneumonia

Rules:

1. $F \land C \rightarrow Flu$
2. $F \land R \rightarrow Measles$
3. $C \land B \rightarrow Pneumonia$

Facts:

F = True, C = True, B = True.

(b) Forward Chaining

Starting facts are Fever, Cough, and Breathlessness.

From Rule 1:

$F \land C \rightarrow Flu$

Since Fever and Cough are true:

Flu = True

From Rule 2:

$F \land R \rightarrow Measles$

Rash is not available, so Measles cannot be derived.

From Rule 3:

$C \land B \rightarrow Pneumonia$

Since Cough and Breathlessness are true:

Pneumonia = True

Result

The possible diagnoses are:

- Flu – True
- Pneumonia – True
- Measles – Not derivable

(c) Backward Chaining for Pneumonia

The goal is:

Pneumonia

The rule producing Pneumonia is:

Cough ∧ Breathlessness → Pneumonia

The system checks whether Cough and Breathlessness are facts. Both are true.

Therefore:

Pneumonia is proved.

CASE STUDY 2 – AUTONOMOUS TRAFFIC MANAGEMENT AGENT

Problem Statement

A smart city traffic controller uses a First-Order Logic knowledge base to manage emergency vehicle routing. It determines whether vehicles can proceed based on emergency status, clear paths, traffic signals, and vehicle positions.

Rules

1. Vehicle(x) ∧ EmergencyType(x) → ClearPath(x)
2. ClearPath(x) → GreenSignal(x) ∧ ¬RedSignal(x)
3. Vehicle(x) ∧ Behind(x,y) ∧ GreenSignal(y) → Proceed(x)

Facts

- Vehicle(Ambulance)
- EmergencyType(Ambulance)
- Vehicle(CarA)
- Behind(CarA, Ambulance)

(a) Unification

Consider Rule 1:

Vehicle(x) ∧ EmergencyType(x) → ClearPath(x)

Match:

Vehicle(x)

with:

Vehicle(Ambulance)

Therefore:

x = Ambulance

The substitution is:

θ = {x/Ambulance}

Since EmergencyType(Ambulance) is also a fact, the rule produces:

ClearPath(Ambulance)

(b) Resolution Proof

Convert Rule 1 into CNF:

¬Vehicle(x) ∨ **¬EmergencyType(x)** ∨ **ClearPath(x)**

Rule 2 gives:

¬ClearPath(x) ∨ **GreenSignal(x)**

and

¬ClearPath(x) ∨ **¬RedSignal(x)**

Rule 3 becomes:

¬Vehicle(x) ∨ **¬Behind(x,y)** ∨ **¬GreenSignal(y)** ∨ **Proceed(x)**

To prove Proceed(CarA), negate the goal:

¬Proceed(CarA)

From the facts:

Vehicle(Ambulance) and EmergencyType(Ambulance)

we derive:

ClearPath(Ambulance)

Then:

GreenSignal(Ambulance)

We already know:

- Vehicle(CarA)
- Behind(CarA, Ambulance)

Therefore Rule 3 gives:

Proceed(CarA)

This contradicts the negated goal ¬Proceed(CarA). Hence the empty clause is obtained.

Conclusion

Proceed(CarA) is logically proved by resolution.

(c) Simultaneous Emergency Vehicles

The rules are general and can handle multiple emergency vehicles because variables are used. However, the current knowledge base contains emergency information only for the ambulance.

Additional facts must be provided for another emergency vehicle, such as:

Vehicle(FireTruck)

EmergencyType(FireTruck)

and its relevant traffic conditions.

Therefore, the FOL structure can support simultaneous emergencies, but the current facts are insufficient to reason about a second emergency vehicle.

CASE STUDY 3 – AGRICULTURAL EXPERT REASONING SYSTEM

Problem Statement

An agricultural advisory agent uses a propositional knowledge base to recommend irrigation methods and determine crop risk.

Knowledge Base

1. SoilDry → IrrigationNeeded
2. IrrigationNeeded ∧ CropWheat → ApplyDripMethod
3. ¬ApplyDripMethod ∧ CropWheat → CropAtRisk

Facts

- SoilDry = True
- CropWheat = True

(a) Conversion into CNF

P1

SoilDry → IrrigationNeeded

After eliminating implication:

¬SoilDry ∨ **IrrigationNeeded**

P2

IrrigationNeeded ∧ CropWheat → ApplyDripMethod

After eliminating implication and applying De Morgan's law:

¬IrrigationNeeded ∨ **¬CropWheat** ∨ **ApplyDripMethod**

P3

¬ApplyDripMethod ∧ CropWheat → CropAtRisk

After conversion:

ApplyDripMethod ∨ **¬CropWheat** ∨ **CropAtRisk**

Facts

SoilDry

CropWheat

Therefore, the complete CNF is:

1. ¬SoilDry ∨ IrrigationNeeded
2. ¬IrrigationNeeded ∨ ¬CropWheat ∨ ApplyDripMethod
3. ApplyDripMethod ∨ ¬CropWheat ∨ CropAtRisk
4. SoilDry
5. CropWheat

(b) Resolution Refutation

Goal:

ApplyDripMethod

Negated goal:

¬ApplyDripMethod

From:

SoilDry

and:

¬SoilDry ∨ IrrigationNeeded

we derive:

IrrigationNeeded

Using:

IrrigationNeeded, CropWheat, and Rule 2:

ApplyDripMethod

Now we have both:

ApplyDripMethod

and:

¬ApplyDripMethod

Therefore, resolution produces the empty clause:

□

Conclusion

ApplyDripMethod is logically entailed by the knowledge base.

(c) Removing SoilDry

If SoilDry is removed:

- IrrigationNeeded cannot be derived.
- ApplyDripMethod cannot be derived.
- CropAtRisk cannot automatically be derived.

This is because absence of ApplyDripMethod does not mean ¬ApplyDripMethod in classical logic.

Therefore, the system must have an explicit negative fact if it wants to conclude CropAtRisk.

CASE STUDY 4 – WUMPUS WORLD KNOWLEDGE AGENT

Problem Statement

An AI agent operates in a Wumpus World environment. It receives percepts such as Stench, Breeze, and Glitter and uses logical inference to determine possible locations of hazards and gold.

Given Percepts

- Stench at [1,2]
- Breeze at [1,1]
- Glitter at [2,2]

Rules

1. Stench[1,2] → Wumpus is adjacent to [1,2]
2. Breeze[1,1] → Pit is adjacent to [1,1]
3. Glitter[x,y] → Gold is at [x,y]
4. ¬Wumpus[x,y] ∧ ¬Pit[x,y] → Safe[x,y]

(a) Belief State Using Modus Ponens

From:

Stench[1,2]

the agent concludes that the Wumpus is adjacent to [1,2].

Possible Wumpus locations are:

- [1,1]
- [1,3]
- [2,2]

From:

Breeze[1,1]

the agent concludes that a Pit is adjacent to [1,1].

Possible Pit locations are:

- [1,2]
- [2,1]

From:

Glitter[2,2]

the agent concludes:

Gold[2,2]

Belief State

Wumpus ∈ {[1,1], [1,3], [2,2]}

Pit ∈ {[1,2], [2,1]}

Gold = [2,2]

(b) Forward Chaining

Starting with Stench[1,2]:

Stench[1,2] → Wumpus adjacent to [1,2]

Starting with Breeze[1,1]:

Breeze[1,1] → Pit adjacent to [1,1]

Starting with Glitter[2,2]:

Glitter[2,2] → Gold[2,2]

Thus, forward chaining allows the agent to derive the possible hazard locations and the location of the gold.

(c) Safety of Path

Proposed path:

[1,1] → [1,2] → [2,2]

At [1,1], Breeze indicates that a Pit may be nearby.

At [1,2], the available knowledge does not establish that the cell is free of hazards.

At [2,2], Glitter confirms the presence of gold, but the knowledge base does not prove that the cell is free from the Wumpus or a Pit.

Therefore, the complete path cannot be logically guaranteed to be safe.

Conclusion

The agent can conclude that:

Gold is at [2,2]

but it cannot prove that:

[1,1] → [1,2] → [2,2]

is completely safe using the available knowledge.

Overall Conclusion

The four case studies demonstrate how logical reasoning techniques are applied in Artificial Intelligence.

The Smart Medical Diagnosis System uses propositional logic, forward chaining, and backward chaining to identify possible diseases. The Autonomous Traffic Management Agent demonstrates First-Order Logic, unification, and resolution for emergency vehicle routing. The Agricultural Expert System demonstrates CNF conversion and resolution refutation to derive irrigation recommendations. The Wumpus World Agent demonstrates Modus Ponens and forward chaining to reason about hazards and gold.

These examples show that knowledge-based AI agents can derive new information from existing facts and rules. However, the reliability of the conclusions depends on the completeness and correctness of the knowledge base.

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